Quadratic Kernel Mapping and Kernel Trick

Vectors:

$a = [a_1, a_2]^T$

$b = [b_1, b_2]^T$

$(a^T b)^2 = a_1^2 b_1^2 + 2a_1 a_2 b_1 b_2 + a_2^2 b_2^2$

$= (a_1 b_1)^2 + 2(a_1 b_1)(a_2 b_2) + (a_2 b_2)^2$

Where: $\varphi(x) \mapsto$

$\varphi(x) = [x_1^2, \sqrt{2}\, x_1 x_2, x_2^2]^T$

$\varphi(a)^T \varphi(b) = a_1^2 b_1^2 + 2a_1 a_2 b_1 b_2 + a_2^2 b_2^2$

$= (a^T b)^2$

2D ⇒ 3D